# The Meaning of the World

| | |
|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 2 seconds |
| Memory limit: | 1024 megabytes |

The meaning of the world must lie outside the world
All things in the world are as they are and happen as they happen
There is no value in the world
—— Tractatus Logico-Philosophicus

Fengmi gives you a string $s$ of length $n$, consisting only of '`(`' and '`)`'.

You need to support $q$ operations:

1. Flip the $k$-th bracket, i.e., change '`(`' to '`)`' and '`)`' to '`(`'.
2. Query the length of the longest valid parentheses substring in the interval $[l, r]$.

Formally, you need to compute:

$$f(l, r) = \max(\{j - i + 1 \mid l \le i \le j \le r \wedge \text{valid}(i, j)\} \cup \{0\})$$

where $\text{valid}(i, j)$ indicates that the substring $s[i..j]$ is a valid parentheses sequence.

In this problem, a valid parentheses sequence is defined as follows:

1. The empty sequence is a valid sequence.
2. If $A$ is a valid sequence, then `(`$A$`)` is also a valid sequence.
3. If $A$ and $B$ are both valid sequences, then $AB$ is also a valid sequence.

## Input

The first line contains two integers $n$ and $q$ ($1 \le n, q \le 2 \times 10^5$) — the length of the bracket string and the number of operations.

The second line contains a string $s$ of length $n$, consisting only of the characters '`(`' and '`)`'. The given string is not guaranteed to be a valid parentheses sequence.

The next $q$ lines each describe one of the following two operations:

- `1 k` ($1 \le k \le n$): flip the $k$-th bracket.
- `2 l r` ($1 \le l \le r \le n$): query the length of the longest valid parentheses substring in the interval $[l, r]$.

## Output

For each operation of type 2, output a single integer on its own line — the answer.

## Examples

| standard input | standard output |
|---|---|
| 8 5<br>()(()())<br>2 1 8<br>1 4<br>2 1 8<br>1 7<br>2 1 8 | 8<br>4<br>4 |
| 12 6<br>((())(()()))<br>2 1 12<br>1 3<br>2 1 12<br>1 8<br>2 1 12<br>2 4 9 | 12<br>10<br>12<br>2 |

## Note

Sample 1:

- Initial string: `()(()())`.
- First query on interval $[1, 8]$: the longest valid parentheses substring is `()(()())` (positions $1 - 8$), length 8.
- Flip the 4th bracket: `()())())`.
- Second query on interval $[1, 8]$: the longest valid parentheses substring is `()()` (positions $1 - 4$), length 4.
- Flip the 7th bracket: `()())(()`.
- Third query on interval $[1, 8]$: the longest valid parentheses substring is `()()` (positions $1 - 4$), length 4.

Sample 2:

- Initial string: `((())(()()))`.
- First query on interval $[1, 12]$: the longest valid parentheses substring is the entire string, length 12.
- Flip the 3rd bracket: `((()()(()()))`.
- Second query on interval $[1, 12]$: the longest valid parentheses substring is `(()()(()())` (positions $3 - 12$), length 10.
- Flip the 8th bracket: `((()()()(()))`.
- Third query on interval $[1, 12]$: the longest valid parentheses substring is `((()()()(()))` (positions $1 - 12$), length 12.
- Fourth query on interval $[4, 9]$: the longest valid parentheses substring is `()` (positions $4 - 5$), length 2.